Lucas de Linhares

lucasdelinhares@gmail.com | lucasdelinhares.com

linkedin.com/in/lucas-de-linhares | github.com/lucascompython

Education

Polytechnic University of Viana do Castelo — Viana do Castelo, PT
Bachelor's Degree in Network and Computer Systems Engineering — *Sep. 2025 - Present*

Polytechnic University of Viana do Castelo — Viana do Castelo, PT
Higher Professional Technical Course in Information Systems Technologies and Programming — *Sep. 2023 - May 2025*

Monserrate Secondary School — Viana do Castelo, PT
Professional Course in Electronics, Automation and Control — *Sep. 2020 - Jul. 2023*

Experience

Development Contributor — Oct. 2026 - Present
MyIPVC - Polytechnic University of Viana do Castelo — *Viana do Castelo, PT*

- Reviewed application security and maintainability, documenting authentication, authorization and data exposure issues with reproduction steps and proposed fixes
- Updated the Node.js backend and its dependencies, reducing vulnerabilities reported by the npm audit from 78 to 0 in the updated version
- Adapted authentication and file storage to the new APIs, preserving mobile application compatibility and fixing functional issues

Research Grant Holder — Jun. 2026 - Aug. 2026
Polytechnic University of Viana do Castelo — *Viana do Castelo, PT*

- Conducted a literature review on Zero Trust Architecture (ZTA) in IT/OT environments within Industry 5.0
- Developed an AI-assisted platform that automates keyword extraction, thematic clustering, synthesis matrix generation, and research gap identification using LLMs
- Designed a parallel processing pipeline with intelligent caching to accelerate literature analysis and reduce redundant LLM processing
- Ongoing collaboration on the preparation of a scientific manuscript after the grant ended.

Software Engineer Intern — Feb. 2025 - Jun. 2025
JCC - Diagnostic Imaging — *Viana do Castelo, PT*

- Developed a company-wide document management and workflow platform using Rust, Actix-Web, Svelte and MySQL
- Designed a fully configurable architecture for users, permissions, document types and workflows
- Implemented RBAC, notifications, due-date tracking, vacation management and internal communication features

Machine Learning Intern — May 2023 - Jul. 2023
Ctrl + Alt DEV — *Viana do Castelo, PT*

- Developed predictive maintenance prototypes for industrial machinery using TensorFlow, Scikit-Learn, Pandas and NumPy, comparing threshold rules, Random Forest and neural networks
- Extracted, filtered and preprocessed real machine data from the MES PostgreSQL database, separating training and test sets for model evaluation
- Developed a FastAPI service to communicate with the MES and display alerts to operators

Software Engineer Intern — May 2022 - Jul. 2022
Ctrl + Alt DEV — *Viana do Castelo, PT*

- Contributed to an industrial Manufacturing Execution System (MES) using Node.js, TypeScript, AdonisJS, React and PostgreSQL
- Investigated PLC integration with the platform, evaluating Siemens S7, OpenPLC and SCADA and testing JavaScript execution triggered by a virtual PLC
- Diagnosed and fixed issues in a Java connector for SAP S/4HANA communication, alongside platform feature development and debugging
- Started developing a Raspberry Pi remote-controlled vehicle, planning Python GPIO control, ultrasonic and proximity sensors and camera-based vision with OpenCV

Projects

Automatic Cocktail Machine (Final Vocational Project) Sep. 2022 - May 2023

Python, Raspberry Pi, PLC, GPIO, HMI, FastAPI, React, 3D Printing

- Integrated a PLC, Raspberry Pi, HMI and motor control into an automatic cocktail preparation machine developed as a team project
- Implemented control logic and GPIO communication between the Raspberry Pi and PLC to coordinate movement and liquid dispensing with peristaltic pumps
- Developed a web interface with React and FastAPI and designed custom mechanical components for 3D printing

Mipster Oct. 2025 - Jan. 2026

Zig, MIPS Assembly, WebAssembly

- Developed a simulator for a subset of the MIPS architecture in Zig, with registers, memory and execution of arithmetic, logical and control flow instructions
- Implemented Assembly parsing and basic system calls, including input and output operations
- Built native and WebAssembly versions with an interactive web editor and syntax highlighting

Photography E-commerce Platform Nov. 2020 - Present

Rust, Axum, SvelteKit, TypeScript, PostgreSQL, Stripe, S3

- Designed, built and continue to maintain a production e-commerce platform handling thousands of monthly transactions
- Developed an optimized media pipeline with AVIF conversion, dynamic watermarking and scalable S3-compatible storage
- Built administration, analytics and sales reporting tools and optimized the frontend and backend to achieve perfect Lighthouse scores

Macro Center Mar. 2023 - Present

Rust, Tauri, Svelte

- Developed a cross-platform application with node-based visual programming and a macro execution engine for keyboard and mouse automation
- Developed libmacrocenter, a Rust library with Python, Node.js, and Lua bindings for desktop automation, powering Macro Center.

Numera Jul. 2026 - Present

Rust, Image Processing, Computer Vision, Parallel Processing

- Developed a Rust application with parallel image processing and configurable print-ready exports, including DPI adjustments, watermarks and overlays
- Developing a computer vision pipeline to recognize athlete and vehicle numbers in event photographs
- Implemented tools to automatically organize thousands of photographs by recognized or assigned numbers

Autism Support Automation Platform Sep. 2023 - Jan. 2024

Go, PostgreSQL, Docker, React, Nginx

- Selected as one of three volunteers for a university accessibility project in the Netherlands
- Diagnosed and resolved Freshdesk access issues, deploying fixes to production
- Analyzed backend code and configuration to identify and fix test failures
- Documented automation workflows and deployment procedures

CritBot Sep. 2020 - Present

Python, PostgreSQL

- Designed and continue to maintain a Discord bot with a modular architecture and a custom internationalization (i18n) system in Portuguese and English
- Integrated external APIs, including Return YouTube Dislike and SponsorBlock, and support for multiple streaming platforms
- Implemented audio features including filters, text-to-speech, automatic music recommendations and media file conversion
- Developed a system for managing text and slash commands with persistent storage in PostgreSQL

Technical Skills

Languages: C, C++, Python, Rust, Zig, TypeScript, SQL, Java, C#, Go
Systems and Automation: PLC, Raspberry Pi, GPIO, HMI, prototyping and 3D printing
Frameworks: React, Next.js, Svelte, FastAPI, Django, Tauri, Axum, Angular, TensorFlow, Stripe
Developer Tools: Git, Docker, GitHub Actions, Netlify, Vercel, Linux, CMake, MySQL, Nginx, PostgreSQL
Spoken Languages: Portuguese (Native), English (Fluent), French (Professional Proficiency)